Lexical Task Heads on Simple Tasks

Schematic project summary

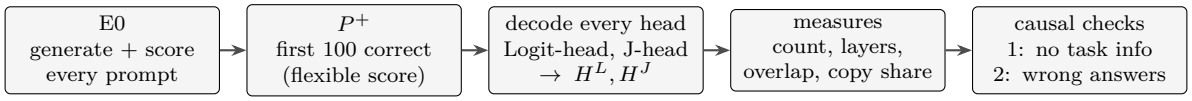
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Marks: ✓ works / supports the hypothesis, ✗ does not, ∼ mixed or task-dependent. Full methods and tables: `proposal-simple.tex`. The complex tasks are summarised in `summary-complex.tex`.

1 Setup in one page

Question	Does the lexical-task-head method of Yang et al. (2026) reproduce on our setup, does Head-JLens (J-head) find heads the per-head logit lens (Logit-head) misses, and which heads should be kept when a task is asked for in several wordings?
Model	Llama-3.1-8B-Instruct (bf16; only model with released Head-JLens matrices).
Readouts	Logit-head: head write $c_{l,h}$ decoded by the unembedding. J-head: head output $z_{l,h}$ mapped through the fitted Jacobian $J_{l,h}$, then decoded.
Simple tasks	14 Todd et al. tasks (antonym, capitals, translation, ...). Rule style: the first 5 of Yang's 6 instruction templates (A–E, incl. the cloze form); example style: 1/5/10/20/30 shots; zero-shot prompt. Raw text, no chat template.
Head criterion	Head is a lexical task head on a prompt if ≥ 1 of its top-10 tokens is in the task's term list; for the style if this holds on $\geq 10\%$ of P^+ (first 100 correct prompts of the most accurate wording).
Term lists	Yang et al.'s (2–22 token ids). Since 11-05 also lists built from the words all heads decode (J-head; E3 below).
Causal check 1	Yang App. G.1: patch the mean head output ($\times\alpha$) at the last token of prompts *without* task info; control = mean of another task at the same heads.
Causal check 2	Yang Sec. 2.4.1/App. H.1.3 in their released script's setting (1-shot): patch prompts that *give* the task but were answered wrongly (baseline 0).
Head sets patched	H^L, H^J, and J-only $H^J \setminus H^L$.



2 Experiments and outcomes

Experiment	Design	Outcome (short)	
E0 — behavioural baseline			
Simple tasks	14 tasks × 11 prompts, every item.	All tasks well above zero-shot; accuracy rises 1→5 shots then flat. Template spread largest on translation, next/prev-item, park–country, synonym.	✓
Scoring audit	Every reply saved; strict (first token) vs flexible.	Translation spread mostly quotes/capitals; first-token score marks any number correct.	✓

E1 — simple tasks (14 tasks × rule/example × 2 readouts)

Experiment	Design	Outcome (short)	
Head sets	100 prompts per style; most accurate template / 5-shot.	Logit-head finds heads on every task. J-head finds more on 23/28 pairs; its extra heads sit in layers 10–19 (Logit-head mostly ≥20). J-head contains all of Logit-head only on 4/28. Example heads mostly inside rule heads (70% shared by Yang's ratio; they report 73%). Counts within 1 head of Yang's Figs. 79/88 in their code's setting; below their Figs. 28/20.	✓
Check 1 (zero-shot)	Means per wording, 15 α values, control mean.	Logit-head: gain ≥10 items on capitals, landmark, park, product (+ currency, next-item, plural with examples). Nothing on antonym, synonym, present–past, prev-item, translations. Best around $\alpha = 4$. Control mostly low (not on park–country, next-item).	∼
Check 1, J-head	same	J-head ≥ Logit-head on most tasks; J-only heads have effects of their own.	✓
Cross-style means	Rule heads × example means and vice versa.	Same tasks gain, often as much or more.	✓
Check 2 (1-shot wrong)	Yang's released script, no control.	Gains on all 14 tasks; reproduces Yang's Figs. 57/62 task order. J-head far stronger on next/prev-item with examples.	✓
E1b — heads from several templates (rule style, simple tasks)			
$k = 5$ (all templates)	Heads passing under all 5 templates; both checks.	Better on fact tasks (capitals, park, product) only; worse elsewhere. Beats E1 on ≤5/14 tasks (≤3 by >3 items).	✗
$k = 1$ (union)	Exploratory.	Helps J-head on present–past (0→16 items), synonym, prev-item, control flat. Hurts Logit-head on fact tasks.	∼
$k = 3$	Exploratory.	Mostly the E1 heads; behaves like E1.	–
$k = 1$, shared	J-head union, kept only heads in the union sets of ≥ 2 tasks (keeps 15–62 of 19–74 heads per task); both checks.	**Not successful.** Removes most of the union's gains: present–past 16→1 items (check 1), 0.64→0.45 (check 2); synonym 0.21→0.13. Only country–capital rises (0.60→0.69). Shared heads do not carry the task better.	✗
$k = 1$, 4 present–past heads	The 4 heads only present–past's union set contains (L19H10, L19H24, L20H28, L21H9; none in E1); both checks, present–past only.	Check 1: no effect alone (0.01); needs the other 15 heads too. Check 2: 0.63, as the whole 19-head union (0.64); rises with α up to 10, real past forms (60/61 distinct replies); $\alpha = 0$ gives 0.02.	∼
Replies, present–past	Each patched reply sorted by kind, per head set and α (J-head, rule means; notebook below).	Shared heads make the model *name* the task ("past tense of assess": 65% of replies in check 1, 73% in check 2 at $\alpha = 10$) without doing it. The 4 unshared heads make it give *a form of the input*: from a bare prompt it repeats the verb (44%), when the prompt shows the task they fix the inflection. Together (union) they give the answer. One task; heuristic reply kinds.	–
Replies, all tasks	Same, every task.	Many patched replies contain the expected answer after a prefix the flexible score rejects ("Capital – Tbilisi", "plural needles", "Country – Russia, . . . "): 19–33% of E1 J-head replies at $\alpha = 4$ on capitals, landmark, antonym, synonym, singular–plural. Not all are answers ("opposite of solid" for *hollow*).	–
Retrieval heads	Yang et al.'s retrieval heads (their code): Definition 1 with the prompt's own answer token as target, on E1's identification prompts; overlap with every lexical set, against layer-matched chance.	Mostly distinct from lexical task heads: 2–9% of each set's heads are retrieval heads, at chance under Logit-head (0.6–1.2×), slightly above under J-head (1.1–1.7×, highest for $k = 5$). None on the translation tasks. Of the 4 present–past unshared heads only L21H9 is one (15% of prompts).	–
4 heads split	Present–past, both checks: L21H9 (the one retrieval head of the 4) alone, and L19H10, L19H24, L20H28 alone.	L21H9 carries the effect: check 2 0.60 alone (4 heads 0.63, 3 heads 0.24); in check 1 it alone makes the model repeat the verb (44% at $\alpha = 10$), as the 4 heads do, and the 3 heads do nothing. A retrieval head that brings the input word forward; the inflection comes from the prompt's example (check 2) or the shared heads (check 1).	∼

Experiment	Design	Outcome (short)	
$k = 1$ without retrieval heads	$k = 1$ sets (L and J) minus every retrieval head of the task (any template, either readout): 51/389 L and 84/672 J heads removed, none on translations; both checks, all tasks.	Mostly small changes, but task-dependent. Landmark–country collapses (check 1: L 0.72→0.02, J 0.75→0.13; J replies still name the country, not first). Present–past J loses its gains (L21H9 removed). Gains on synonym J (0.21→0.32), product–company J (0.35→0.48, control 0.10) and prev-item J check 2 (0.12→0.45, at α 9–10, no constant answer). Park–country L: control now above task. Unchanged sets reproduce $k = 1$ exactly. Broad removal: retrieval heads of E1's template and the same readout only would remove 12 L and 51 J heads.	∼
Equal-size sets	Top $N \in \{5, 10, 20\}$ heads by match rate (ties: term count) under each readout and rule (E1, $k = 1/3/5$), only heads matching ≥ 1 prompt (37/420 sets not formed); both checks, own-style means, 9 α $(-10, -4, 2\text{–}6, 8, 10)$; threshold sets compared on the same 9.	**Readouts: J-head's lead is not set size.** Of 25 threshold-set J leads ≥ 0.05, 16 hold at every N, none reverse; none of Logit-head's 4 leads holds (country–capital, present–past reverse). **Rules: mostly size.** Most E1-vs-k differences flip with N; consensus ($k = 5$) below E1 in check 2 (J: antonym, singular–plural, next-item, German). No steady gain with N. Present–past J example: threshold (4 heads) 0.00, top 5 0.87 by adding L15H1 (rate 0.07), control 0; only sets with L15H1 move check 1. L15H1 alone (example means): 0.80 / 0.76 in checks 1 / 2 (top 5: 0.87 / 0.80; the 4 without it 0.00 / 0.57), control 0, none with rule means; J-head reads past participles from it. Control high on park–country (title starting with the country), singular–plural J ex $N = 5$, next-item J ex $N = 20$.	∼
Davidson FV	Davidson et al.'s (2025) instruction function vector, 4 tasks (landmark, product, English–French, present–past): heads ranked by the gain in answer probability when set to their mean on prompts whose instruction is replaced (equiprobable tokens, WikiText, other tasks' instructions), top 20 over the 4 tasks; FV = sum of their mean outputs (inst1–4, 100 prompts), added after layer 11 at the last prompt token; our check 1, 15 α ($\alpha = 1$ is theirs), control.	Peak at $\alpha = 1$ everywhere, nothing at $\lvert\alpha\rvert \geq 3$. Landmark 0.85, product 0.64 (controls 0.06 / 0.12; our best sets 0.79 / 0.46 at $\alpha \approx 3$–4); English–French 0.09 (control 0.03; ours $\leq$0.02); present–past 0.02 (repeats the verb, as our sets). FV heads in layers 13–17 mostly, 0–5 shared with each E1 set. English–French: 8 of the 9 hits are the English input copied where the French differs only in accents. **80 heads** (next 60 by the same score; FV 25–30% longer, cosine 0.88–0.91 with the 20-head FV): still best at $\alpha = 1$; landmark 0.96, product 0.61, English–French 0.19 (11 real translations), present–past 0.07 (real past forms); controls $\leq$0.10. **Set size** (5–1,024 heads, same ranking): no single best. Facts: 5–10 heads at α 3–4 best (landmark 0.95, product 0.87; product control 0.22), at $\alpha = 1$ 40–80 heads (0.95–0.96 / 0.61). English–French: 40 heads, $\alpha = 1$, 0.28 (0.20 real translations). Present–past: 320 heads, $\alpha = 1$, 0.18 (real past forms). All 1,024 heads: facts break (0.01). Size chosen on the test items.	∼
Davidson FV, 14 tasks	As above, heads chosen over all 14 tasks (30 heads; next_item: 12 and 28 correct prompts on two templates, so fewer for its mean); our check 1, all 14 tasks.	Best at $\alpha = 1$ nearly everywhere; FV $\geq$ our best E1 set on 11/14 tasks, controls low ($\leq$0.12). Large gains where our sets do little: singular–plural 0.60 (ours $\leq$0.03), antonym 0.26 ($\leq$0.04), park–country 0.89 (0.39–0.48, our controls 0.29–0.32), English–Spanish 0.10 (real translations). Also above: capitals 0.83 (0.70), landmark 0.89, product 0.64, currency 0.13. Nothing on present–past, prev-item, synonym, English–German (0.01–0.04); English–French 0.09 = copied inputs (accents ignored). Next-item 0.10 vs J 0.17 (J control 0.35).	∼

E3 — head sets from decoded words (J-head only, rule style; check 1 only, rule means averaged over the 5 templates, control mean, $\alpha \in \{1, 2, 4, 8\}$, best α reported. Numbers are means over the 14 tasks; E1 J-head on the same α: 0.19.) Words: a decoded top-10 token, stripped and lowercased, kept if alphabetic, ≥ 3 letters, not a stop word, common English (`wordfreq` Zipf ≥ 3); *share* = fraction of (prompt, head) pairs over all 1,024 heads whose top-10 holds it; *task words* = share in the task − mean share in the other 13; *common words* = highest minimum share over the 14 tasks. Heads: Definition 1 with the new list ($k = 10$, $\rho = 0.10$) unless stated. Only P^+ (correct) prompts.

Experiment	Design	Outcome (short)	
Top decoded words	Top 40 words per task over all heads, no baseline.	Task names and answer types lead (*opposite*, *city*, *currency*, *French*); generic words on every task (*what*, *which*, *maybe*, *https*). Yang's list words only partly among them.	–
Common + task, 4 tasks	20 common and 20 task words from the best template; heads on the best template; both readouts; antonym, present–past, capitals, product.	Task list J best (capitals 0.59, antonym 0.22); common lists ≈ 0 except product under Logit-head (0.32).	~
All 14 tasks	Same, words counted on all 5 templates.	Task J 0.10, common J 0.07: below E1.	✗
List length	5, 10, 15, 20 words.	Shorter is better: task J 0.22 (5 words) $\rightarrow$ 0.10 (20). Common ≤ 0.07.	~
Heads on pooled prompts	Definition 1 on the $\approx$500 prompts of the 5 templates pooled instead of the best template; also Yang's list.	Helps every list: task J 5 words 0.25; common J 0.11–0.17 (synonym 0.58 with 10 common words, control high); Yang's list pooled 0.21.	~
Combined 5+5 (2×2)	5 task + 5 common words; words and heads each from the best template or pooled.	**Both pooled = CC-pooled**: 0.45, 95–137 heads, above E1 on all 14 tasks; carries tasks E1 does not (antonym 0.54, singular–plural 0.64, next-item 0.43, translations 0.25–0.32). Present–past 0.03. Controls: synonym 0.26, park 0.17, singular–plural 0.12, next-item 0.12. Words pooled, heads best 0.30; words best, heads pooled 0.31; both best 0.18.	✓
Random sets	Best set per task so far vs 5 layer-matched random sets of its size.	Best 0.49, random 0.03–0.07. Single draws high on antonym (0.45) and landmark (0.42); product 0.16–0.39 on all draws.	✓
Strict consensus	5+5 and 10+10 lists, head must pass on all 5 templates, top-5 tokens, $\rho = 0.15$ (run at $\rho = 0.05$ stopped); also its top 20 by rate.	0.15–0.20. Best only on country–capital (0.78).	✗
Neutral cut (CC-pooled-clean)	J-head readings on 1,000 WikiText-103 passages (lengths matched); the 50 words heads read most there removed before choosing task and common words; 5+5 and 10+10, pooled.	5+5: 0.38, 10+10: 0.33. Above CC-pooled on 8/14 (facts, translations, morphology: landmark 0.97, French 0.40, German 0.49, singular–plural 0.72, present–past 0.14). Loses antonym, synonym, next/prev-item: they need the generic words. New common words: *named, therefore, question, answer, very.*	~
Task minus WikiText	Top 10 task words after the cut; no cross-task contrast, no common words.	0.21. Landmark 0.00 (patched replies start "Country"), translations name the language.	✗
Group contrast	5 groups (word relations, morphology, facts, sequences, translations); task words = share − max share of the other tasks of its group; + 5 top WikiText words (*which, mph, http, his, what*).	0.32. Best antonym of all (0.70). Landmark and park collapse (0.01 / 0.08): *country* is shared within the group and removed.	~
Match-rate ranking	CC-pooled heads ranked by pooled match rate; top 1, 2, 5, 10, 20, 40; layer-matched random at 10, 20.	Small sets work only on capitals (L16H25 alone 0.88) and landmark (40 heads 0.93). Means: 5 heads 0.19, 20 heads 0.21, 40 heads 0.25.	~
Relation heads	Head's top-10 holds both a term-list token and a token of the prompt's own query.	Rare: $\rho \geq 0.10$ for ≤ 7 heads per task, none on 10 tasks. Not patched. L16H25 decodes *city*, not the relation.	–
Attend + decode	L16H25 pattern: at the last token it attends to *capital*, *country* in the instruction and decodes *city*. Score = (share of last-token attention on task words in the instruction, outside the query) $\times$ (share of prompts with a task word in the J-head top 10); task words = Yang's list $\cup$ 10 top task words; eager attention, pooled prompts. Top 1, 2, 3, 5, 10, 20; layer-matched random at 5, 10.	Few heads carry fact tasks and plurals: capitals 0.88 (L16H25 alone), landmark 0.79 (5 heads), product–company 0.72 (20 heads, control 0.09; best of all), singular–plural 0.68 (20). Antonym 0.25 (5). Nothing on translations, next/prev-item, present–past. Park: control nearly as high. L16H25 ranks first on all 5 fact tasks; pattern in layers 13–17. Means: 5 heads 0.21, 20 heads 0.25 (match rate 0.19 / 0.21); random 0.04.	~

3 Hypotheses

H1 replication	Fails with the zero-shot patching of Yang App. G.1 (check 1: no gain on 7 tasks); the G.1 figures reproduce only with check 2 (wrong prompts, Sec. 2.4.1/App. H.1.3), which gains on all 14.	∼
H2 J-head ⊇ Logit-head	Fails: J-head misses part of the Logit-head set on 24/28 pairs. J-only heads are causal in both checks.	✗
H3 all-template heads	Falsified: better on ≤5/14 tasks.	✗

4 Design variants tried

Date	Tried first	Replaced by	Why / effect
10-27	Rules as main style, examples secondary	Rule and example treated alike	No main condition (user decision).
10-28	Prompts and means pooled over wordings	One wording for heads, one mean per wording (Yang's code)	Matches the source exactly.
10-28	Yang's batching without attention mask	Padding masked	Model attended to padding.
10-28	Check 1 only (zero-shot)	Check 1 + check 2 (released script)	Only check 2 reproduces Yang's figures.
10-30	First-token (simple) / strict (complex) score for heads and checks	Flexible score; E1, E2 rerun	Strict scoring created spurious wording differences.
11-02	E1b with $k = 5$ only	$k \in \{1, 3, 5\}$	After $k = 5$ failed; exploratory.
11-05	Yang's hand-written term lists	Lists from decoded words (E3), J-head only	Exploratory; light check 1 (α 1, 2, 4, 8).
11-05	Words and heads from the best template; 20-word lists	Both pooled over 5 templates; 5 + 5 words (CC-pooled)	0.18 → 0.45; shorter lists better.
11-05	Strict consensus at $\rho = 0.05$	$\rho = 0.15$	Run stopped and restarted (user decision).
11-05	Common words from task prompts	Words generic text elicits removed first (WikiText)	Helps facts, translations; hurts antonym, synonym, sequences.
11-06	Small sets by match rate	By attention to task words × decoding	User's L16H25 observation; larger effects at 5 and 20 heads.

Inspecting replies: `notebooks/patching_outputs.py` (marimo) shows the simple-task patched replies for every manipulation: task, check, procedure (E1, k = 1/3/5, shared/unshared), readout, head and mean style, mean source and wording, α; accuracy, the share of the most common reply, reply kinds, and each item's replies.

5 What seems to work, what does not

- ✓ **Scoring matters first.** Flexible scoring removes most of the apparent wording sensitivity; always check it before reading wording effects.
- ✓ **Check 2 (wrong prompts, released-script setting)** is what reproduces Yang et al.'s figures, including those App. G.1 attributes to zero-shot patching; the zero-shot check works only on fact-like tasks.
- ✓ **J-head on simple tasks:** more heads, earlier layers (10–19), and the added heads carry the task causally. It does not contain the Logit-head set.
- ∼ **Which heads to keep** depends on the task type: all-template heads for fact tasks, union of templates for morphology-like tasks under J-head.
- ∼ **Retrieval heads** (heads that already write the answer token) are few among lexical task heads, but some effects rest on them: landmark–country's on its 10, present–past's union gain on L21H9. Elsewhere removing them changes little or helps (synonym, product–company, prev-item under J-head). Entity-answer tasks may be carried by retrieval more than by task heads.
- ∼ **Two roles on present–past** (one task): heads shared across tasks make the model *name* the task; L21H9 makes it give *a form of the input*. Either alone fails from a bare prompt; with an example in the prompt, L21H9 alone suffices.
- ✓ **J-head vs Logit-head at equal size:** J-head's larger effects mostly remain with 5, 10 or 20 heads per readout; Logit-head's few leads do not. Differences between template rules mostly come from set size.

- ✓ **Words the heads actually decode make better lists** (J-head, check 1): CC-pooled (5 task + 5 common words, words and heads over all 5 templates) is above E1 on all 14 tasks (mean 0.45 vs 0.19) and carries antonym, singular–plural, next-item and the translations, where E1 does nothing. Both word kinds and the pooling are needed.
- ∼ **Generic words help some tasks.** Removing the words heads read on any text (WikiText) helps facts, translations and morphology but hurts antonym, synonym and sequences.
- ∼ **Few heads suffice for named answer types:** heads that attend to the instruction's task words and decode the answer type carry capitals (1 head), landmark (5), product–company and singular–plural (20). Not translations, sequences, present–past.
- **E3 caveats:** words, list sizes and designs chosen after seeing check 1 on the same items; best of 4 α; one run; no check 2, no held-out items yet.
- **Read the replies, not only accuracy:** 19–33% of patched replies on several tasks contain the answer after a prefix the flexible score rejects, and part of a "loss" can be a format change (landmark–country under J-head). `notebooks/patching_outputs.py` shows them.

Not pursued: function vectors, function-specific heads, hidden-state differences (scope, 10-27); fp32 Head-JLens fit (≈136 GPU-h, declined).